

MCQs:-

1. B

reason:- because science is the scientific knowledge & engineering making things from technology and chemistry is the branch of science.

2. D

reason:- other ~~to~~ options just doesn't make any sense.

3. A

reason:- the next options also linked with quantum mechanical model.

4. C

reason:- the other options doesn't make any sense.

5. A

reason:- The statement ~~ask~~ ^{doesn't} ask for ionization energy, electronegativity and shielding effect.

6. C

8. B

9. C

7. D

10. A

Q.2(i)

(a) Saturated solution :-

A saturated solution is a solution that contains the maximum amount of solute that can dissolve in a given amount of solvent at a particular temperature.

(b) Unsaturated solution :-

An unsaturated solution is a solution that contains less than the maximum amount of solute that can dissolve at a particular temperature.

(c) Increasing temperature increases the solubility of NH_4Cl , but decreases the solubility of Ca(OH)_2 .

Q.2(ii)

(a)

$$Ar = \frac{(79 \times 50.5)}{100} + \frac{(81 \times 49.5)}{100}$$

$$= \frac{3989.5}{100} + \frac{4009.5}{100}$$

$$= \frac{7999}{100} = 79.99$$

(b) Protons:- 35
 Electrons :- 35
 neutrons = $81 - 35$
 $= 46$

(c) They have the same number and arrangement of electrons, especially the same number of outer shell electrons.

6.2(iii)

(a) X:-

highest shell = 3 $\rightarrow$ Period:- 3rd
 outer electrons = 2 $\rightarrow$ group:- ~~8+2~~ 10
 $= 10$

Y:-

Period:- 3rd
 group:- $6 + 10$
 $= 16$

(b) X has 2 outer electrons, so it loses 2:-
 X^{2+}

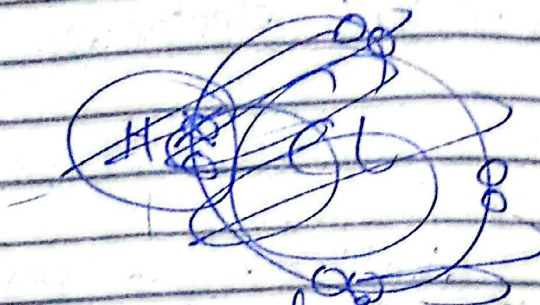
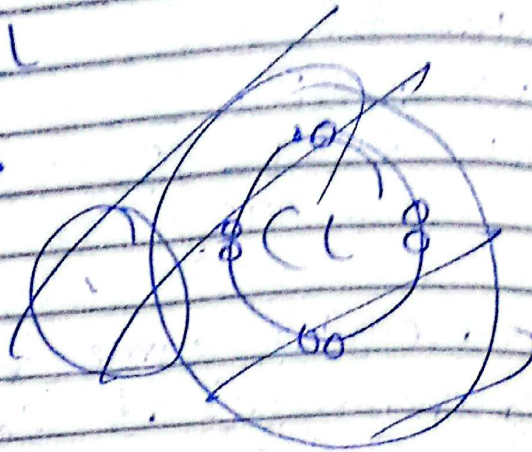
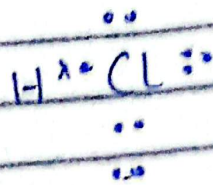
Y needs 2 electrons to complete its outer shell:-

Y^{2-}

(c) XY^{2-}

Q: 2iv

HCl



(b) The H-Cl bond is polar because Cl is more electronegative than H, so the bonding electrons are attracted more strongly towards chlorine.

Q: 2(v)

(i) Cu

$$n + (-2) = 0$$

$$n - 2 = 0$$

$$n = +2$$

(ii) Cu

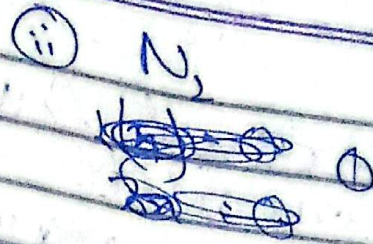
$$n = 0$$

$$\textcircled{b} \textcircled{ii} \text{NH}_3$$

$$n + 3(1) = 0$$

$$n - 3 = 0$$

$$n = +3$$



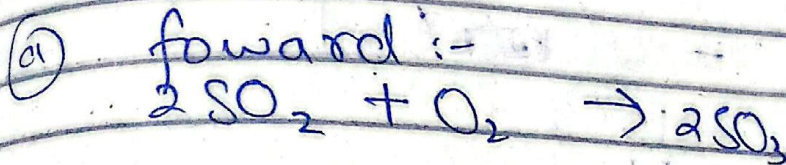
$$\textcircled{c} \text{ Nitrogen} = +3 \rightarrow 0$$

$$\text{Copper} = +2 \rightarrow 0$$

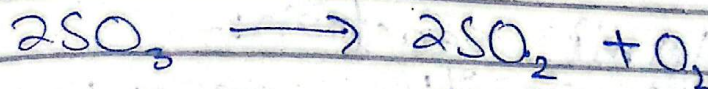
$\left. \begin{array}{l} \text{oxidized} \\ \text{reduced} \end{array} \right\}$

oxidizing agent = CuO
 reducing agent = NH₃

Q: 2(vi)



reverse :-



b) At the very start no SO₃ has been formed, so SO₃ available to react in the reverse direction.

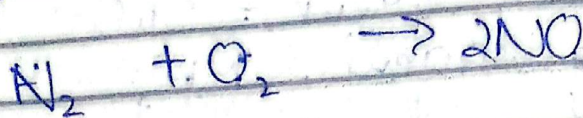
Therefore :-

Reverse reaction rate = 0

(i) Rate of forward reaction is equal to the rate of reverse reaction.

(ii) Concentration of reactants and products remain constant.

Q.3(i)



$$[945\text{ kJ} + 498\text{ kJ}] \Rightarrow 2[945 + 498]$$

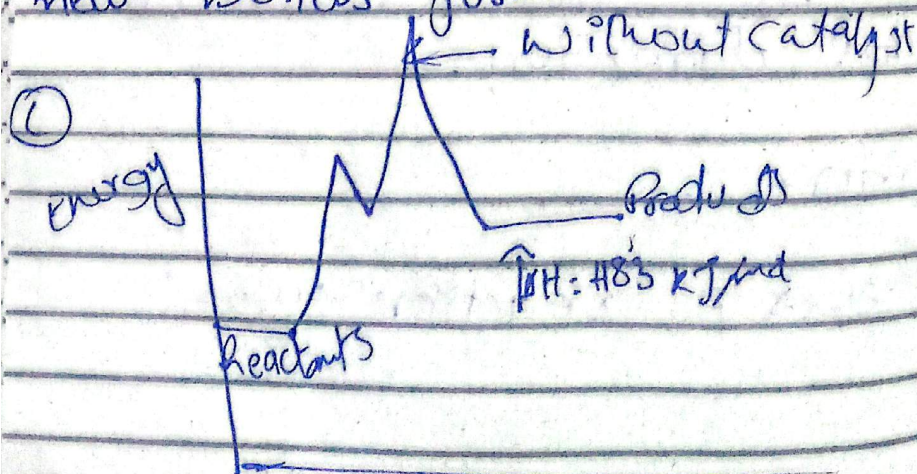
$$1443 - 2(1260)$$

$$1443 - 1260$$

$$\Delta H = +183 \text{ kJ/mol}$$

(b) Endothermic reaction:-

ΔH is positive, meaning more energy is break bonds than is released when new bonds form.



① Catalyst :-
Changes :-

Activation energy - it lowers E_a

Does not change :- Enthalpy change
 ΔH

Q:3(ii)

① $n = \frac{\text{mass}}{\text{Relative At. M}}$

$$n(C) = \frac{1.8}{12} = 0.15 \text{ mol}$$

$$n(H) = \frac{0.3}{1} = 0.30 \text{ mol}$$

Divide by smallest value :-

$$C : \frac{0.15}{0.15} = 1$$

$$H = \frac{0.30}{0.15} = 2$$

$$= 1:2$$

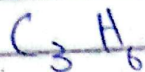
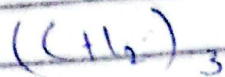


(b) Empirical formula mass is

$$12 + (6 \times 1) = 18$$

$$Mr = 42$$

$$\frac{42}{18} = 3$$



(c) moles in 8.4 g

$$n = \frac{m}{Mr}$$

$$= \frac{8.4}{42}$$

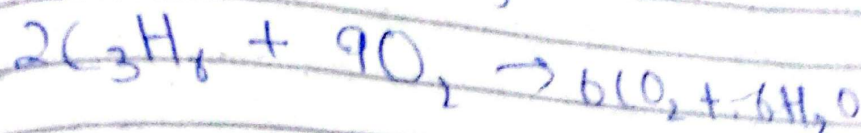
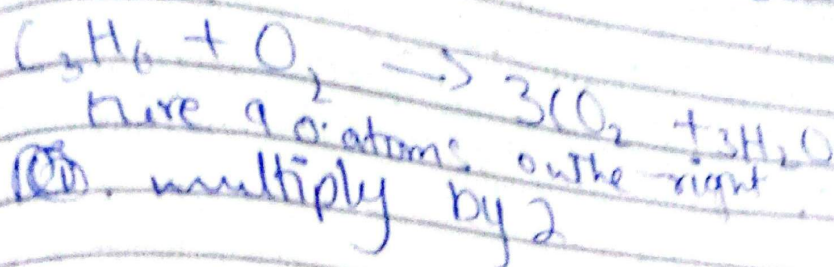
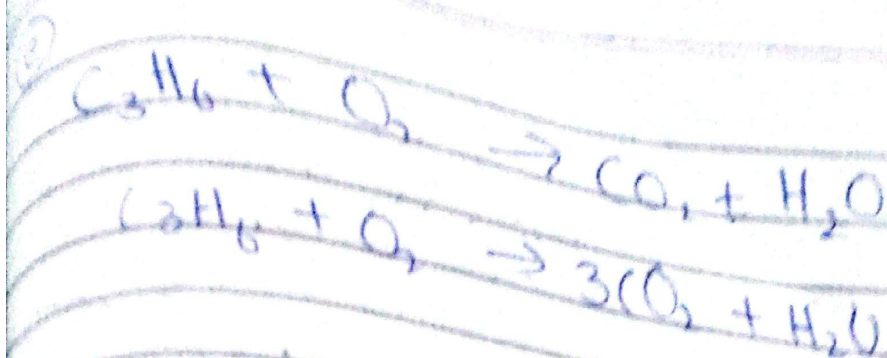
$$= 0.20 \text{ mol}$$

(d) N.o of molecules in 8.4 g

$$N = n \times N_A$$

$$= 0.20 \times 6.022 \times 10^{23}$$

$$= 1.204 \times 10^{23} \text{ molecules}$$



① m.m of $P = 42 \text{ g/mol}$

1 mole contain $= 6.022 \times 10^{23}$

$$= \frac{42}{6.022 \times 10^{23}}$$

mass of one molecule $= 6.98 \times 10^{-23} \text{ g}$